

Jesús Ruiz, Erik Alonso, Elisabete Aramendi, Jo Kramer-Johansen, Trygve Eftestøl, Unai Ayala, Digna González-Otero. Reliable extraction of the circulation component in the thoracic impedance measured by defibrillation pads. *Resuscitation*, Volume 84(10), 2013, Pages 1345-1352, ISSN 0300-9572, <https://doi.org/10.1016/j.resuscitation.2013.05.020>.

(<https://www.sciencedirect.com/science/article/pii/S0300957213003018>)

Abstract

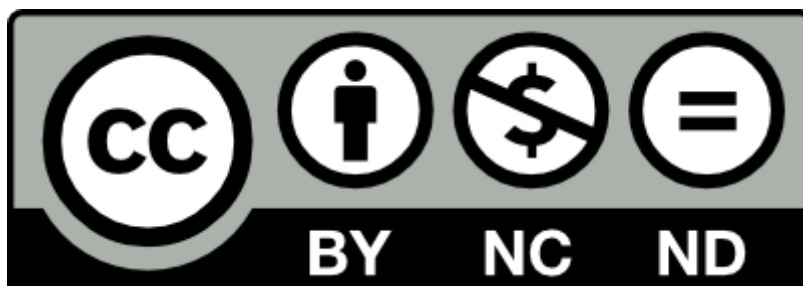
Aim: To analyze the feasibility of extracting the circulation component from the thoracic impedance acquired by defibrillation pads. The impedance circulation component (ICC) would permit detection of pulse-generating rhythms (PRs) during the analysis intervals of an automated external defibrillator when a non-shockable rhythm with QRS complexes is detected.

Methods: A dataset of 399 segments, 165 associated with PR and 234 with pulseless electrical activity (PEA) rhythms, was extracted from out-of-hospital cardiac arrest episodes by applying a conservative criterion. Records consisted of the electrocardiogram and the thoracic impedance signals free of artifacts due to thoracic compressions and ventilations. The impedance was processed using an adaptive scheme based on a least mean square algorithm to extract the ICC. Waveform features of the ICC signal and its first derivative were used to discriminate PR from PEA rhythms.

Results: The segments were split into development (83 PR and 117 PEA rhythms) and testing (82 PR and 117 PEA rhythms) subsets with a mean duration of 10.6s. Three waveform features, peak-to-peak amplitude, mean power, and mean area were defined for the ICC signal and its first derivative. The discriminative power in terms of area under the curve with the testing dataset was 0.968, 0.971, and 0.969, respectively, when applied to the ICC signal, and 0.974, 0.988 and 0.988, respectively, with its first derivative.

Conclusion: A reliable method to extract the ICC of the thoracic impedance is feasible. Waveform features of the ICC or its first derivative show a high discriminative power to differentiate PR from PEA rhythms (area under the curve higher than 0.96 for any feature).

Keywords: Automated external defibrillator (AED); Circulation detection; Pulse-generating rhythm (PR); Pulseless electrical activity (PEA); Thoracic impedance.



Reliable extraction of the circulation component in the thoracic impedance measured by defibrillation pads

J Ruiz^a, E Alonso^{*,a}, E Aramendi^a, J Kramer-Johansen^b, T Eftestøl^c, U Ayala^a, D
González-Otero^a

Affiliation and addresses:

^a Communications Engineering Department.
University of the Basque Country UPV/EHU.
Alameda Urquijo S/N
48013 Bilbao, Spain

^b Institute for Experimental Medical Research.
Oslo University Hospital and University of Oslo.
N-0424 Oslo, Norway

^c Department of Electrical Engineering and Computer Science.
University of Stavanger.
4036 Stavanger, Norway

Corresponding author:

^{*} Erik Alonso
email: erik_alonso@ehu.es
Tel. : +34946017384
Fax. : +34946014259

Word counts:

Abstract: 260 words

Paper : 2867 words

Abstract

Aim: To analyze the feasibility of extracting the circulation component from the thoracic impedance (TI) acquired by defibrillation pads. The impedance circulation component (ICC) would permit detection of pulse-generating rhythms (PR) during the analysis intervals of an automated external defibrillator when a non-shockable rhythm with QRS complexes is detected.

Methods: A dataset of 399 segments, 165 associated with PR and 234 with pulseless electrical activity (PEA) rhythms, was extracted from out-of-hospital cardiac arrest episodes by applying a conservative criterion. Records consisted of the electrocardiogram and the TI signals free of artifacts due to thoracic compressions and ventilations. The impedance was processed using an adaptive scheme based on a least mean square (LMS) algorithm to extract the ICC. Morphological features of the ICC signal and its first derivative were used to discriminate PR from PEA rhythms.

Results: The segments were split into development (83 PR and 117 PEA rhythms) and testing (82 PR and 117 PEA rhythms) subsets with a mean duration of 10.6 s. Three morphological features, peak to peak amplitude, mean power, and mean area were defined for the ICC signal and its first derivative. The discriminative power in terms of area under the curve (AUC) with the testing dataset was 0.968, 0.971, and 0.969, respectively, when applied to the ICC signal, and 0.974, 0.988 and 0.988, respectively, with its first derivative.

Conclusion: A reliable method to extract the ICC of the TI is feasible. Morphological features of the ICC or its first derivative show a high discriminative power to differentiate PR from PEA rhythms (AUC>0.96 for any feature).

Keywords

Thoracic impedance, Automated External Defibrillator (AED), Pulse-generating Rhythm (PR), Pulseless Electrical Activity (PEA), Circulation detection.

1. INTRODUCTION

Detecting signs of circulation is essential in interventions for cardiopulmonary resuscitation (CPR), especially when either lay people or professionals use defibrillators. Up until 1998, the European Resuscitation Council guidelines required pulse checking to determine cardiac arrest in basic life support.¹ Since 2000, the carotid pulse has not been included in the protocol,² as it was proved to be both time consuming and inaccurate.^{3,4} The 2010 guidelines established recognition of cardiopulmonary arrest based on unresponsiveness and abnormal breathing.⁵ When using an automated external defibrillator (AED), they recommend resuming immediate CPR after the shock is provided or after no-shock is indicated. The rescuer should watch continuously for the responsiveness of the patient looking for movement and normal breathing.

In that context, an automated circulation detection method integrated with the shock advice algorithm (SAA) of the AED would help the rescuer to recognize signs of circulation. The SAA analyzes the cardiac rhythm in artifact-free intervals, with no chest compressions or ventilations. When the SAA detects a non-shockable rhythm with the presence of QRS complexes, an automated circulation detector would discriminate pulseless electrical activity (PEA) rhythms, with no circulation, from pulse-generating rhythms (PR).

Transthoracic-impedance plethysmography measured with four electrodes has been used for decades as a non-invasive technique for estimating stroke volume and cardiac output.⁶ The thoracic impedance (TI) provides information regarding the blood circulation; pulse-generating heartbeats produce small temporary changes in impedance,⁷ and a close relationship between changes in TI and variable aortic pressures has been shown.⁸ Current AEDs measure the TI continuously through self-adhesive defibrillation pads. Therefore, it has been suggested that the AEDs should be expanded to include the TI as a potential hemodynamic sensor.⁹⁻¹³

The TI signal recorded through defibrillation pads may show different components: 1) a baseline impedance of 50-120 Ω depending on the patient size and body composition; 2) fluctuations due to thoracic compressions and ventilations between 0.2 and several ohms; 3) an impedance circulation component (ICC) with fluctuations less than 100 m Ω in pulse-generating rhythms; and 4) other noise and artifacts due to movement, electrode-skin contact, and so on. The intervals for rhythm analysis in the AED are free of compressions and ventilations. In these intervals, the ICC, if present, could be detected provided there is an effective noise removal. Reliable extraction of the ICC is

31 key to detecting the presence of the pulse automatically. The magnitude of the ICC is smaller in
32 PEA rhythms than in PR due to the lack of circulation in PEA rhythms. The hypothesis of this
33 study is that a reliable method can be developed to extract the circulation component from the
34 impedance recorded through defibrillation pads. The method could be integrated into an AED
35 system for automated pulse detection during analysis intervals.

2. MATERIALS AND METHODS

2.1. Data materials

The data were obtained from a prospective study of out-of-hospital cardiac arrest (OHCA) episodes, recorded between March 2002 and September 2004 in Akershus, Stockholm, and London.^{14,15} The study was designed to measure the quality of out-of-hospital CPR performed by ambulance personnel in adherence to the resuscitation guidelines of 2000. A modified version of the HeartStart 4000 defibrillators (Philips Medical Systems, Andover, MA, USA) was used to record the following signals with a sampling rate of 500 Hz: the surface ECG (resolution 1.03 μV per least significant bit with bandwidth 0–50 Hz), the TI (resolution 0.74 m Ω per least significant bit with bandwidth 0–80 Hz) by applying a sinusoidal excitation current (32 kHz, 3 mA peak to peak) between the defibrillation pads, and additional reference channels. Expert reviewers annotated the episodes using five rhythm types: ventricular fibrillation (VF) and ventricular tachycardia (VT) in the shockable category, and asystole (AS), PEA, and PR in the non-shockable category. PR intervals were annotated either when a pulse was detected clinically or when TI changes simultaneous with cardiac contractions were detected.¹⁴

Retrospectively, we extracted segments from those episodes with 4–15 s duration, according to the typical length for the analysis pauses in an AED, and with the TI signal free of artifacts due to chest compressions or ventilations. To avoid uncertainty in the PR/PEA classification of the extracted segments, a restrictive criterion was applied in the extraction: the PEA segments were extracted from episodes with a complete absence of PR annotations along the episode, and the PR segments were selected from the final interval of the episodes annotated with return of spontaneous circulation and with the patient admitted alive to hospital. Several segments from the same patient were considered if they corresponded to separate stages of the intervention. Each record consisted of two signals, the ECG and the TI, both downsampled to $f_s=250$ Hz.

The database was split randomly into two datasets for development and testing, each with a similar number of patients and segments. When a patient contributed with more than one segment they were separated into the two datasets. Table 1 presents a summary of the composition of the database which comprised 399 segments from 127 patients.

2.2. Adaptive extraction of the ICC signal from the TI signal

The ECG signal, $ecg[n]$, was processed to compute the instantaneous frequency of the QRS complexes, $f_i[n]$. Then the ICC signal, $icc[n]$, was extracted from the TI signal.

ECG signal processing

As shown in Fig 1, the ECG signal was band-pass filtered at 0.5–30 Hz to suppress the baseline drift and the high-frequency artifacts. QRS complexes were annotated automatically using an offline Matlab version of the QRS detector running in the commercial monitor-defibrillator Reanibex 800 (BexenCardio, Ermua, Spain). The detector is compliant with the standard IEC¹⁶ and reported sensitivity and positive predictive values above 99.5% when tested on the MIT-BIH database. Instants of the QRS complexes, n_i $i=1..M$, were computed for the M complexes of the segment. The $f_i[n]$ was computed as:

$$f_i[n] = \frac{1}{(n_{i+1} - n_i)} \quad n_i \leq n < n_{i+1} \quad \text{for } i = 1..M - 1, \quad (1)$$

where n_i and n_{i+1} are the instants in samples of two consecutive QRS complexes.

Fig 2 shows the ECG signal, $ecg[n]$, for a PR segment (panel a) and for a PEA segment (panel b). Red dashed lines depict the instants of the QRS complexes.

TI signal processing

The TI signal, $z[n]$, was preprocessed as shown in Fig 1. First, the direct component (DC) due to the baseline impedance was removed by subtracting the mean value. Then, the high-frequency noise was suppressed using a low-pass filter with a f_{c1} cut-off frequency. Finally, the signal was high-pass filtered with a f_{c2} cut-off frequency to suppress low-frequency components below the fundamental frequency of the circulation component. Fig 2 shows the raw impedance signal, $z[n]$, and the preprocessed signal, $z_p[n]$, for a PR segment (panel a) and a PEA segment (panel b).

The $z_p[n]$ signal consisted of the ICC and additional noisy components that were removed using an adaptive scheme that incorporates a least mean square (LMS) algorithm.¹⁷ Previous knowledge on spectral analysis of the TI circulation component in hemodynamically stable volunteers¹⁸ showed that the circulation component could be modeled adequately with the first three harmonics. A model based on Fourier coefficients with time-varying amplitude and phase was applied as follows:

$$\hat{icc}[n] = \sum_{k=1}^3 A_k[n] \cos(2\pi k f_i n + \varphi_k[n]) = \sum_{k=1}^3 \hat{a}_k[n] \cos(2\pi k f_i n) + \hat{b}_k[n] \sin(2\pi k f_i n). \quad (2)$$

Fig 1 shows the three stages of the adaptive scheme, each dedicated to computing one of the first three harmonic components of $f_i[n]$ in the ICC. The LMS algorithm recursively computes the amplitudes of the in-phase and quadrature components, $\hat{a}_k[n]$ and $\hat{b}_k[n]$, for $k = 1, 2$, and 3 to estimate the circulation component of the TI, $\hat{icc}[n]$. Each of the three filters that compose the adaptive scheme requires a step-size parameter, μ_k , which defines the adaptive process of the LMS algorithm. For each harmonic component, it describes the convergence rate and the bandwidth of the filter for that component. To reduce the number of step-size parameters, the influence of each harmonic was modeled with decreasing step size for increasing harmonic number:

$$\mu_k = \frac{1}{k} \mu_0 \quad k = 1, 2, 3, \quad (3)$$

where μ_0 is the step size for the fundamental component, $k = 1$.

The development dataset was used to set the optimum values for μ_0 , f_{c1} and f_{c2} which maximize the difference of the ICC estimated for the PR and for the PEA rhythm classes. The μ_0 value was set to $\mu_0 = 0.1\pi/f_s$, and the f_{c1} and f_{c2} cut-off frequencies to 7 Hz and 0.65 Hz, respectively.

Fig 2 shows the estimated circulation component, $\hat{icc}[n]$, in $\text{m}\Omega$, and its first derivative, $\hat{dicc}[n]$, in $\text{m}\Omega \cdot \text{s}^{-1}$, for a PR and a PEA rhythm. A smaller magnitude of the component for the PEA rhythm sample can be observed. For comparison, the typical non adaptive processing for the impedance was also implemented to compute the impedance cardiogram.^{12,13} The first derivative was computed directly for the preprocessed impedance, $z_p[n]$. Panels c and d in Fig 2 show the results for the PR and the PEA rhythm cases, respectively.

2.3. Morphological features

To evaluate the reliability of the method to extract the ICC, three morphological features were defined for the waveforms of signals $\hat{icc}[n]$ and $\hat{dicc}[n]$: peak to peak amplitude, mean power, and mean area.

Peak to peak amplitude: z_1 and dz_1

113 The maximum and minimum amplitudes of the waveform between consecutive QRS complexes
 114 were calculated: max_i and min_i for $\hat{icc}[n]$, and, $dmax_i$ and $dmin_i$ for $\hat{dicc}[n]$. The differences were
 115 computed for every complex as:

$$pp_i = max_i - min_i \quad dpp_i = dmax_i - dmin_i \quad \text{for } i = 1 \dots M - 1, \quad (4)$$

116 The mean peak to peak amplitudes, z_1 and dz_1 , were defined as the mean values of pp_i and dpp_i
 117 for the M-1 RR intervals of the segment:

$$z_1(\text{m}\Omega) = \frac{1}{M-1} \sum_{i=1}^{M-1} pp_i \quad dz_1(\text{m}\Omega \cdot \text{s}^{-1}) = \frac{1}{M-1} \sum_{i=1}^{M-1} dpp_i, \quad (5)$$

118 *Mean power: z_2 and dz_2*

119 The area per sample under the squared of the signals $\hat{icc}[n]$ and $\hat{dicc}[n]$ was computed as:

$$z_2(\text{m}\Omega^2) = \frac{1}{N} \sum_{n=1}^N \hat{icc}[n]^2 \quad dz_2(\text{m}\Omega^2 \cdot \text{s}^{-2}) = \frac{1}{N} \sum_{n=1}^N \hat{dicc}[n]^2, \quad (6)$$

120 where N is the number of samples of the segment.

121 *Mean area: z_3 and dz_3*

122 The mean area is defined as the area per sample under the absolute value of the signals $\hat{icc}[n]$
 123 and $\hat{dicc}[n]$:

$$z_3(\text{m}\Omega) = \frac{1}{N} \sum_{n=1}^N |\hat{icc}[n]| \quad dz_3(\text{m}\Omega \cdot \text{s}^{-1}) = \frac{1}{N} \sum_{n=1}^N |\hat{dicc}[n]|. \quad (7)$$

124 2.4. Evaluation and statistical analysis

125 The reliability of the method used to extract the ICC was evaluated through the morphological
 126 features, attending to their distributions for PR and PEA rhythm groups and to their discriminative
 127 power.

128 Results are presented as median (25-75 percentiles) because their distributions did not pass the
 129 one-sample Kolmogorov-Smirnov normality test. The Wilcoxon rank sum test was used for testing
 130 significant differences between the median values of the variables in different groups. A p-value <
 131 0.05 was considered significant.

132 The reliability of the extracted signals, $\hat{icc}[n]$ and $\hat{dicc}[n]$, was evaluated through receiver
133 operating characteristic (ROC) analysis in terms of the discriminative power of the features
134 measuring the area under the curve (AUC) for the testing dataset.

135 3. RESULTS

136 Table 1 summarizes the characteristics of the database with segments grouped into development
137 and testing datasets. In the PEA rhythm group, 234 segments were gathered (117 development,
138 117 testing) with a median duration of 12.2 s. In the PR group, 165 segments (83 development,
139 82 testing) had a median duration of 7.1 s. The median duration of the segments for the complete
140 database was 10.6 s (7.5-14.1).

141 Table 2 shows the values of median, upper, and lower quartiles of the features for the PR and
142 PEA rhythm groups, along with the significance value of the statistical test and the AUC for the
143 development and testing datasets. The performance for the six features is very similar. It can be
144 observed that the values associated with the PR samples are significantly higher ($p < 0.001$) than
145 the values associated with PEA rhythm samples. The discriminative power is very high, with
146 $AUC > 0.96$ for any feature. Fig 3 shows the ROC graph for the testing dataset. Panel a shows the
147 curves for the z_1 , z_2 , and z_3 features extracted from the $\hat{icc}[n]$. Panel b shows the curves for the
148 dz_1 , dz_2 , and dz_3 features extracted from the $\hat{dicc}[n]$.

149 Fig 4 illustrates the performance of the method with several samples. Panels a and b show a
150 PR and a PEA rhythm segment, respectively. Although their heart rate is similar (81 and 77 bpm)
151 the magnitudes of $\hat{icc}[n]$ and $\hat{dicc}[n]$ for the PR case are higher than for the PEA rhythm case.
152 Panels c and d show examples of where the ICC extracted by the method is abnormal: it is small
153 in a segment annotated as PR (panel c) and high in a PEA rhythm segment (panel d).

4. DISCUSSION

Detection of an absence of pulse in resuscitation scenarios is critical to improve the response to a cardiac arrest. Nevertheless, carotid pulse checking has been reported to be time consuming (median of 24 s) and inaccurate, with 90% sensitivity and 55% specificity.^{3,4,19} Checking for movement, breathing, or coughing (signs of circulation) has not been proven to be superior diagnostically.^{20,21} In this context, monitoring the presence of cardiac activity could be performed by integrating a circulation detector into an AED. When the SAA detects a non-shockable organized rhythm, with the presence of QRS complexes, an automated circulation detector would discriminate PR from PEA rhythms using the TI recorded by the AED.

In contrast to the transthoracic-impedance plethysmography measured with four electrodes, the TI signal recorded through defibrillation pads hides the component reflecting cardiac activity under important noise and artifacts components. Fig 2 shows in detail the first derivative of the $z_p[n]$ signal computed to obtain the impedance cardiogram for a PR segment (panel c) and for a PEA rhythm segment (panel d). It can be observed that the waveforms of the first derivative do not discriminate between the two cases as clearly as the outputs of our method, signals $\hat{icc}[n]$ and $\hat{dicc}[n]$ in panels a and b respectively. Therefore, when dealing with the impedance recorded by defibrillation pads, it is necessary to extract the hidden ICC. This study focuses on the feasibility of extracting the ICC in such a reliable way that any morphological feature will discriminate highly between PR and PEA rhythms.

Losert et al.¹⁰ and Risdal et al.¹¹ addressed the detection of pulse in the AED context using the ECG and the TI signals recorded through defibrillation pads. Losert et al.¹⁰ proposed a set of 9 features of the impedance waveform to discriminate between segments with systolic arterial pressures > 80 mmHg from those with ≤ 80 mmHg. When tested with episodes with mechanical ventilation, the best single parameter showed an AUC of 0.75. Risdal et al.¹¹ performed a rigorous analysis of up to 17 features and defined a neural network classifier with 12 parameters: 7 extracted from the ECG and 5 extracted from the TI. When tested to discriminate PRs from PEA rhythms with in-hospital and out-of-hospital episodes, the best single feature was one extracted from the impedance, which showed an AUC of 0.78 with segments including ventilation artifacts. In both works, they filtered (fixed filters) and averaged the TI signal to remove ventilation and to extract the features. The segments of our study had no compression or ventilation artifacts, according to

the characteristics of the analysis intervals in an AED. The adaptive method we applied extracts a noise-free ICC. Consequently, any of the morphological features of the ICC or its first derivative showed a high discriminative power. The AUC value was superior to 0.96 for any single feature with the testing dataset.

More recently, Cromie et al.^{12,13} addressed the detection of cardiac arrest through the impedance recorded by defibrillation pads in order to identify circulatory collapse accurately. In the first work, an experimental animal study, they discriminated sinus rhythm from VF, PEA, and asystole using the fast Fourier transform of the impedance cardiogram. They reported sensitivity/specificity values of 86%/90%, respectively, with the testing set (sensitivity was defined as the capacity to identify arrest rhythms correctly and specificity was defined as the capacity to identify PR correctly). In the second work, a broad clinical study, they discriminated arrest rhythms (VF, pulseless VT, agonal, PEA, and other pulseless rhythms) from nonarrest rhythms with a global sensitivity/specificity of 81%/97%, respectively, with the testing set. In our scheme, the shock advice algorithm of the AED discriminates accurately between shockable rhythms (arrest rhythms) and non-shockable rhythms. We propose the use of the impedance to discriminate between PEA rhythms and PRs when a non-shockable organized rhythm, with QRS complexes, is detected. The discriminative power obtained is high, as shown in the ROC graphs, with $AUC > 0.96$, which exceeds the 78.6% sensitivity reported by Cromie et al.¹³ for PEA rhythms.

In both works by Cromie et al.,^{12,13} the segments were free of compressions and ventilations. They obtained the impedance cardiogram after filtering the impedance signal at 1–11 Hz, and chose the peak value of the spectrum in the 1.5–4.5 Hz bandwidth as the most sensitive feature. They measured the spectral component of the impedance cardiogram at the main cardiac frequency. In our method, not only the main frequency component at the cardiac rate, but also its second and third harmonics were extracted adaptively, obtaining an ICC with high morphological discriminative power.

The proposed method could be integrated easily into current AEDs for automated pulse detection. Both, ECG and TI signals should be processed in the analysis pauses when the shock advice algorithm detects a non-shockable organized rhythm. The processing required for TI is fast and simple. It includes temporal processing of the TI with fixed filters for preprocessing and a rapid adaptive algorithm, the LMS algorithm, to extract the ICC.

4.1. Limitations of the study

Characteristics of the segments in the databases match characteristics of the analysis intervals in AEDs: mean duration of 10.6 s, no compressions, and no ventilations. Extracted records did not always correspond to the precise instant when the SAA performed the analysis, but they all had the mentioned characteristics.

5. Conclusions

The TI acquired through defibrillator pads shows signs of circulation of the patient, and can be used to identify pulse-generating rhythms in the analysis interval of an AED when a non-shockable rhythm is detected. The component associated with the cardiac activity in the TI is hidden in noise and other frequency components. In this study, an adaptive method is proposed to process the TI and extract the ICC. Morphological features of the ICC signal and its first derivative showed a high discriminative power. When tested with segments extracted from OHCA episodes, any of the features could discriminate PRs from PEA rhythms with $AUC > 0.96$.

Conflict of interest

Authors, J. Ruiz, and E. Aramendi, have received research support from Bexen Cardio (Ermua, Spain) for studies on AED shock advice algorithms.

Acknowledgements

This work received financial support from the Ministerio de Economía y Competitividad of Spain through the projects TEC2012-31928 and TEC2012-31144, from the University of the Basque Country (UPV/EHU) through the unit UFI11/16 and from the Programa de Formación de Personal Investigador del Departamento de Educación, Universidades e Investigación del Gobierno Vasco through the grants BFI-2010-174, BFI-2010-235 and BFI-2011-166.

References

- [1] Bossaert L, Handley A, Marsden A, et al. European Resuscitation Council guidelines for the use of automated external defibrillators by EMS providers and first responders: A statement from the Early Defibrillation Task Force, with contributions from the Working Groups on Basic and Advanced Life Support, and approved by the Executive Committee. *Resuscitation* 1998;37(2):91–94.
- [2] Monsieurs KG, Handley AJ, Bossaert LL. European Resuscitation Council Guidelines 2000 for Automated External Defibrillation: A statement from the Basic Life Support and Automated External Defibrillation Working Group and approved by the Executive Committee of the European Resuscitation Council. *Resuscitation* 2001;48(3):207–209.
- [3] Bahr J, Klingler H, Panzer W, Rode H, Kettler D. Skills of lay people in checking the carotid pulse. *Resuscitation* 1997;35(1):23–26.
- [4] Eberle B, Dick WF, Schneider T, Wisser G, Doetsch S, Tzanova I. Checking the carotid pulse check: diagnostic accuracy of first responders in patients with and without a pulse. *Resuscitation* 1996;33(2):107–116.
- [5] Koster RW, Baubin MA, Bossaert LL, et al. European Resuscitation Council Guidelines for Resuscitation 2010 Section 2. Adult basic life support and use of automated external defibrillators. *Resuscitation* 2010; 81(10):1277–1292.
- [6] Penney BC. Theory and cardiac applications of electrical impedance measurements. *Critical reviews in biomedical engineering* 1986;13(3):227–281.
- [7] Geddes LA, Baker LE. *Principles of Applied Biomedical Instrumentation*. Wiley-Interscience, 1989, 3rd edition.
- [8] Pellis T, Bisera J, Tang W, Weil MH. Expanding automatic external defibrillators to include automated detection of cardiac, respiratory, and cardiorespiratory arrest. *Critical care medicine* 2002;30(4 Suppl):S176–178.
- [9] Johnston P, Imam Z, Dempsey G, Anderson J, Adgey A. The transthoracic impedance cardiogram is a potential haemodynamic sensor for an automated external defibrillator. *European Heart Journal* 1998;19(12):1879–1888.
- [10] Losert H, Risdal M, Sterz F, et al. Thoracic-impedance changes measured via defibrillator pads can monitor signs of circulation. *Resuscitation* 2007;73(2):221–228.
- [11] Risdal M, Aase SO, Kramer-Johansen J, Eftestøl T. Automatic identification of return of spontaneous circulation during cardiopulmonary resuscitation. *IEEE transactions on bio-medical engineering* 2008;55(1):60–68.
- [12] Cromie NA, Allen JD, Turner C, Anderson JM, Adgey AAJ. The impedance cardiogram recorded through two electrocardiogram/defibrillator pads as a determinant of cardiac arrest during experimental studies. *Critical care medicine* 2008;36(5):1578–1584. PMID: 18434896.
- [13] Cromie NA, Allen JD, Navarro C, Turner C, Anderson JM, Adgey AAJ. Assessment of the impedance cardiogram recorded by an automated external defibrillator during clinical cardiac arrest. *Critical care medicine* 2010;38(2):510–517. PMID: 19864942.
- [14] Wik L, Kramer-Johansen J, Myklebust H, et al. Quality of cardiopulmonary resuscitation during out-of-hospital cardiac arrest. *JAMA: the journal of the American Medical Association* 2005;293(3):299–304.
- [15] Kramer-Johansen J, Myklebust H, Wik L, et al. Quality of out-of-hospital cardiopulmonary resuscitation with real time automated feedback: a prospective interventional study. *Resuscitation* 2006;71(3):283–292. PMID:

17070980.

- [16] INTERNATIONAL STANDARD IEC 60601-2-27. Particular requirements for the basic safety and essential performance of electrocardiographic monitoring equipment. Edition 3.0. 2011.
- [17] Widrow B, Stearns SD. Adaptive signal processing. Prentice-Hall, 1985.
- [18] Alonso E, Aramendi E, Ruiz J, Ayala U, González-Otero D. Suppression of the respiration artefact and extraction of the cardiac component in the thoracic impedance recorded through defibrillation pads. *Computing in Cardiology* 2012;.
- [19] Ochoa F, Ramalle-Gómara E, Carpintero J, García A, Saralegui I. Competence of health professionals to check the carotid pulse. *Resuscitation* 1998;37(3):173–175.
- [20] Ruppert M, Reith MW, Widmann JH, et al. Checking for breathing: evaluation of the diagnostic capability of emergency medical services personnel, physicians, medical students, and medical laypersons. *Annals of emergency medicine* 1999;34(6):720–729.
- [21] Perkins GD, Stephenson B, Hulme J, Monsieurs KG. Birmingham assessment of breathing study (BABS). *Resuscitation* 2005;64(1):109–113.

Figure Legends

- Figure 1 Diagram of the processing scheme for the ECG signal (upper panel) and the TI signal (lower panel). The ECG signal, $ecg[n]$, is processed to extract the instantaneous frequency of the QRS complexes, $f_i[n]$. The impedance signal, $z[n]$, is preprocessed to obtain $z_p[n]$ and an adaptive scheme is applied to estimate the ICC signal, $\hat{icc}[n]$.
- Figure 2 Examples of a PR segment and a PEA segment processed with the adaptive scheme proposed in this study (panels a and b, respectively) and processed with a classical non adaptive scheme (panels c and d, respectively). From top to bottom: the ECG signal, $ecg[n]$, in mV; the raw impedance, $z[n]$, in Ω ; the preprocessed impedance, $z_p[n]$, in $m\Omega$; the estimated ICC signal, $\hat{icc}[n]$, in $m\Omega$; and its first derivative, $d\hat{icc}[n]$, in $m\Omega \cdot s^{-1}$. The magnitudes of the last two signals for the PR case are greater than for the PEA rhythm case. Panels c and d show the impedance cardiogram, $dz_p[n]$, in $m\Omega \cdot s^{-1}$, computed directly as the first derivative of $z_p[n]$. Dashed red lines depict the instants of QRS complexes.
- Figure 3 ROC graphs for the morphological features extracted from $\hat{icc}[n]$ and $d\hat{icc}[n]$ signals (panels a and b respectively) for the testing dataset.
- Figure 4 Examples of PR and PEA segments. Panels a and b show two cases with similar heart rates (77 and 81 bpm), but the ICC and its first derivative for the PR case (panel a) are greater than for the PEA case (panel b). Panels c and d show two abnormalities: a PR case (panel c) with small $\hat{icc}[n]$ and $d\hat{icc}[n]$ signals, and a PEA sample (panel d) with signals of magnitudes similar to the PR case.

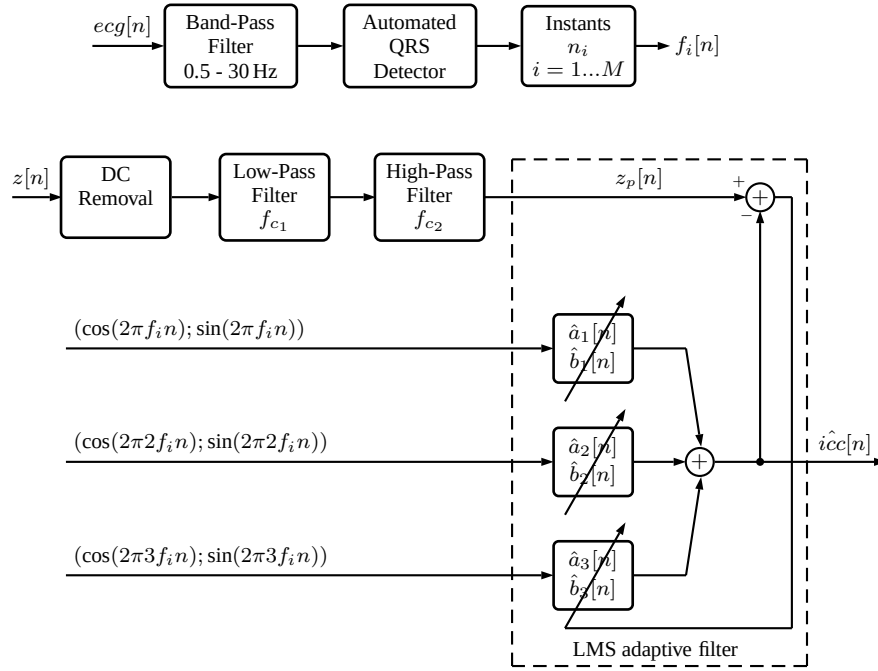


Figure 1: Diagram of the processing scheme for the ECG signal (upper panel) and the TI signal (lower panel). The ECG signal, $ecg[n]$, is processed to extract the instantaneous frequency of the QRS complexes, $f_i[n]$. The impedance signal, $z[n]$, is preprocessed to obtain $z_p[n]$ and an adaptive scheme is applied to estimate the ICC signal, $\hat{icc}[n]$.

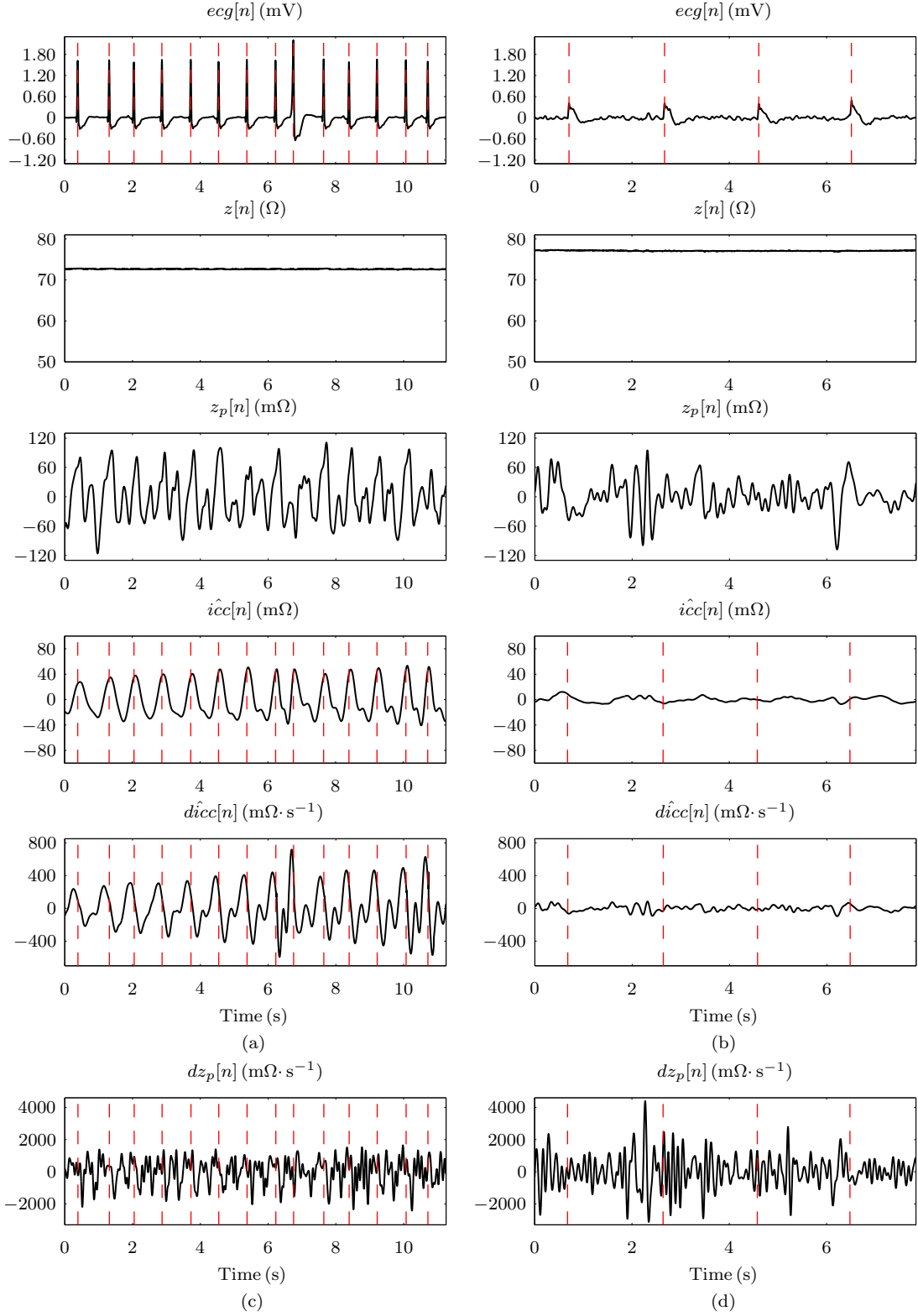


Figure 2: Examples of a PR segment and a PEA segment processed with the adaptive scheme proposed in this study (panels a and b, respectively) and processed with a classical non adaptive scheme (panels c and d, respectively). From top to bottom: the ECG signal, $ecg[n]$, in mV; the raw impedance, $z[n]$, in Ω ; the preprocessed impedance, $z_p[n]$, in m Ω ; the estimated ICC signal, $\hat{icc}[n]$, in m Ω ; and its first derivative, $d\hat{icc}[n]$, in m $\Omega \cdot s^{-1}$. The magnitudes of the last two signals for the PR case are greater than for the PEA rhythm case. Panels c and d show the impedance cardiogram, $dz_p[n]$, in m $\Omega \cdot s^{-1}$, computed directly as the first derivative of $z_p[n]$. Dashed red lines depict the instants of QRS complexes.

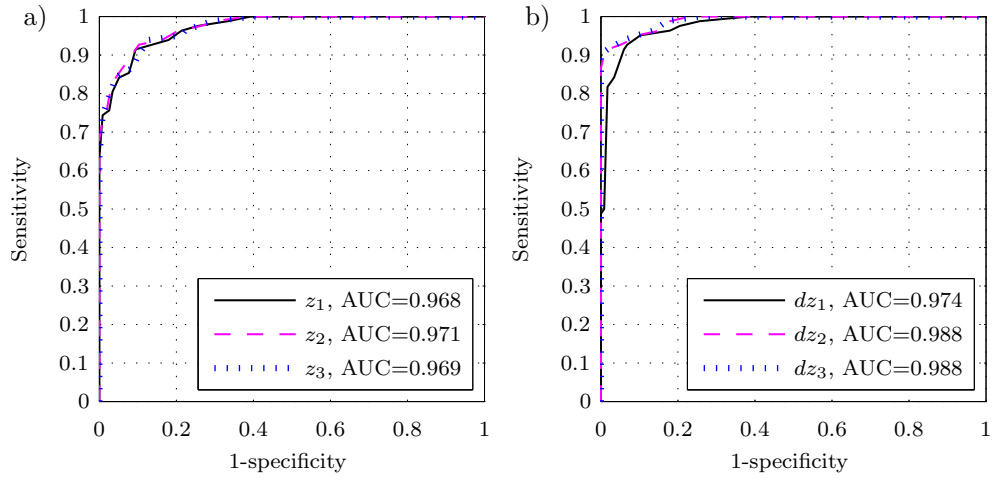


Figure 3: ROC graphs for the morphological features extracted from $\hat{icc}[n]$ and $\hat{dice}[n]$ signals (panels a and b respectively) for the testing dataset.

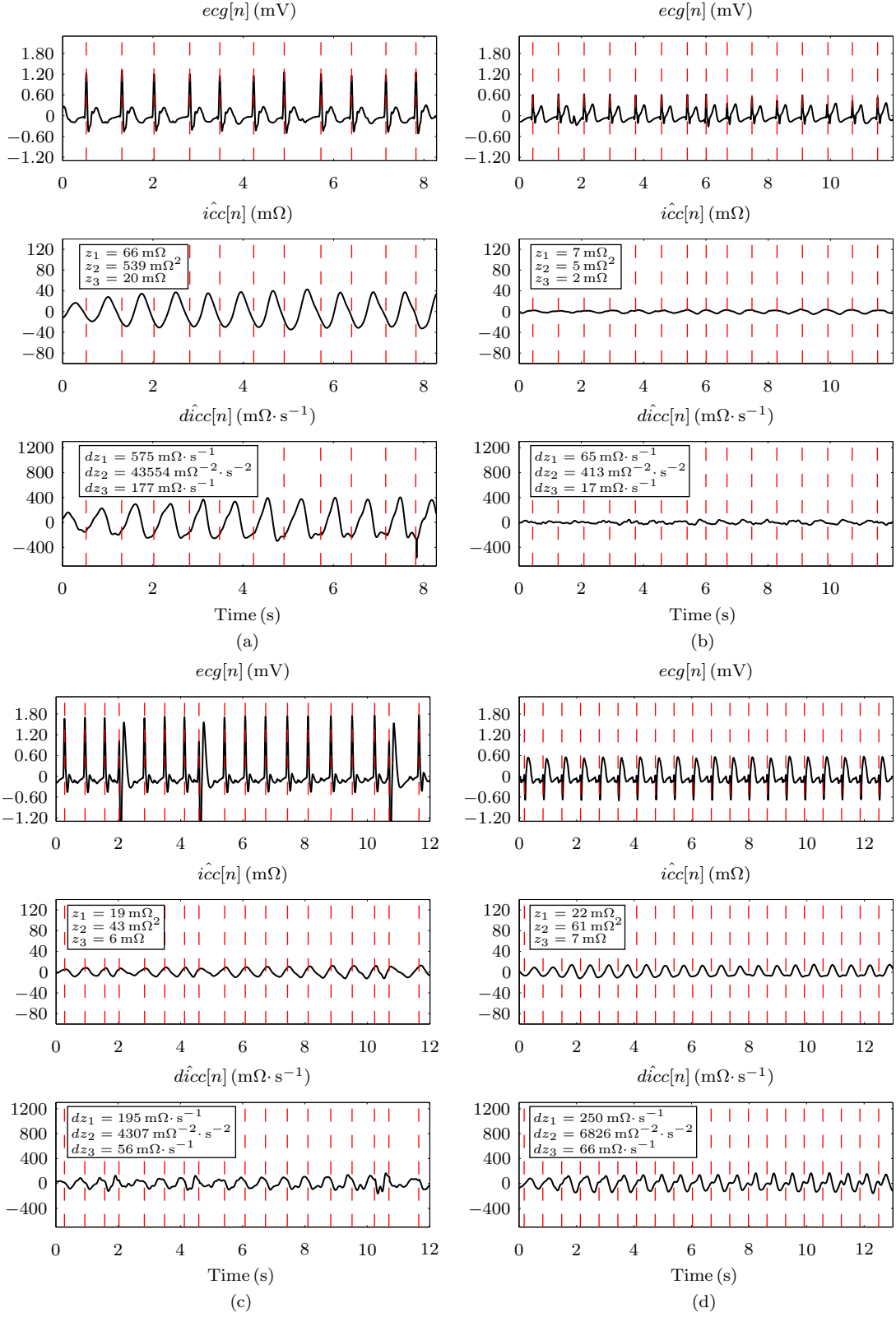


Figure 4: Examples of PR and PEA segments. Panels a and b show two cases with similar heart rates (77 and 81 bpm), but the ICC and its first derivative for the PR case (panel a) are greater than for the PEA case (panel b). Panels c and d show two abnormalities: a PR case (panel c) with small $\hat{icc}[n]$ and $\hat{dicc}[n]$ signals, and a PEA sample (panel d) with signals of magnitudes similar to the PR case.

312 Table Legends

313	Table 1	Summary of segments grouped in development and testing datasets. The
314		number of patients is indicated in parentheses. Duration of the segments
315		is expressed as median (25-75 percentiles).
316	Table 2	Distributions for the six features extracted from the ICC and its first
317		derivative for the development and testing datasets. The median, upper,
318		and lower quartiles of the features are given for PR and PEA rhythm
319		groups, along with the significance value of the statistical test and the
320		AUC.

Table 1: Summary of segments grouped in development and testing datasets. The number of patients is indicated in parentheses. Duration of the segments is expressed as median (25-75 percentiles).

Database	PR		PEA	
	No. segments	Duration (s)	No. segments	Duration (s)
Development	83(40)	7.1(5.3, 11.3)	117(66)	12.3(10.3, 15)
Testing	82(43)	7.0(5.3, 10.3)	117(70)	12.1(9.7, 15)
Total	165(48)	7.1(5.3, 11.2)	234(79)	12.2(9.8, 15)

Table 2: Distributions for the six features extracted from the ICC and its first derivative for the development and testing datasets. The median, upper, and lower quartiles of the features are given for PR and PEA rhythm groups, along with the significance value of the statistical test and the AUC.

Feature	Development				Testing			
	PR	PEA	p-value	AUC	PR	PEA	p-value	AUC
$z_1(\text{m}\Omega)$	40.04 (29.11, 57.72)	6.03 (4.18, 10.81)	<0.001	0.972	46.12 (27.13, 74.06)	6.22 (4.36, 11.11)	<0.001	0.968
$dz_1(\text{m}\Omega \cdot \text{s}^{-1})$	478.85 (328.15, 725.28)	61.61 (48.14, 100.51)	<0.001	0.984	509.45 (368.33, 796.40)	64.90 (48.84, 113.22)	<0.001	0.974
$z_2(\text{m}\Omega^2)$	230.03 (104.73, 473.59)	3.90 (1.57, 12.94)	<0.001	0.974	298.61 (88.67, 694.34)	4.08 (1.83, 14.60)	<0.001	0.971
$dz_2(\text{m}\Omega^2 \cdot \text{s}^{-2})$	30164 (11686, 63525)	266.19 (131.98, 872.06)	<0.001	0.991	29167 (15833, 98963)	301.95 (143.65, 948.50)	<0.001	0.988
$z_3(\text{m}\Omega)$	12.54 (8.14, 18.10)	1.57 (1.02, 2.94)	<0.001	0.975	14.25 (8.18, 21.97)	1.61 (1.10, 3.27)	<0.001	0.969
$dz_3(\text{m}\Omega \cdot \text{s}^{-1})$	138.99 (86.98, 204.04)	12.84 (8.63, 24.03)	<0.001	0.991	136.59 (97.45, 242.56)	13.66 (9.35, 24.32)	<0.001	0.988